GG-R-1638A
18 December 1985
SUPERSEDING
GG-R-001638 (DM)
9 April 1970

FEDERAL SPECIFICATION

REMOVER, DENTAL CROWN (Simplex)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a crown remover suitable for use during dental operative procedures.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitations for bid or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

GG-I-526	Instruments, Dental and Surgical, General Specifications.
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-676	Boxes, Set-up.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

(Copies of listed federal and military standards, specifications, Commercial Item Descriptions (CIDs), handbooks and associated documents listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting officer.)

(Federal Government activities may obtain copies of Federal Standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM E18	Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E92	Tests for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The crown remover shall be in accordance with the applicable requirements of GG-I-526, with the options and additional requirements as specified herein.

3.2 Material.

3.2.1 Shaft and Points. The shaft shall be fabricated of Class 3, Type 303 or Class 4, Type 416 or 440A, stainless steel. The points shall be fabricated of Class 4, Type 440A, stainless steel.

3.2.2 Weight, sleeve and cap. The weight, sleeve, and cap shall be fabricated of chrome-plated brass. The weight, sleeve, and cap shall pass the boil test specified in GG-I-526.

3.3 Hardness. After proper stress relieving, the points shall have a Rockwell hardness of C52 to C56 when tested as specified in 4.4.1.

3.4 Style, design and dimensions. The crown remover shall be of the type known as "Simplex", and shall be suitable for use during dental operative procedures. The crown remover shall be furnished with one contra-angle point (Type 1) and one right-angle point (Type 2). The points shall be of the cone socket type, and shall be readily interchangeable into the distal end of the shaft. The style, design, and dimensions of the crown remover shall be in accordance with figure 1.

3.4.1 Figure. Figure 1, forming a part of this specification, is a specification control drawing. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions on tolerances is permissible without prior approval of the contracting officer.

3.5 Construction. The crown remover shall consist of a cap, shaft, weight, sleeve, and two interchangeable points. The cap shall be knurled, and shall be securely and permanently attached to the proximal end of the shaft except the alternate cap shall be removable by unthreading. The weight shall move freely along the shaft between the cap and the sleeve (stops). The weight shall weigh 53 $\pm$ 3 grams. The threaded, knurled sleeve shall be removable from the distal end of the shaft to allow for the removal of the weight. The cap shall withstand the impact test without failure, when tested as specified in 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 20260-26T11:54Z
Check the source to verify that this is the current version before use.

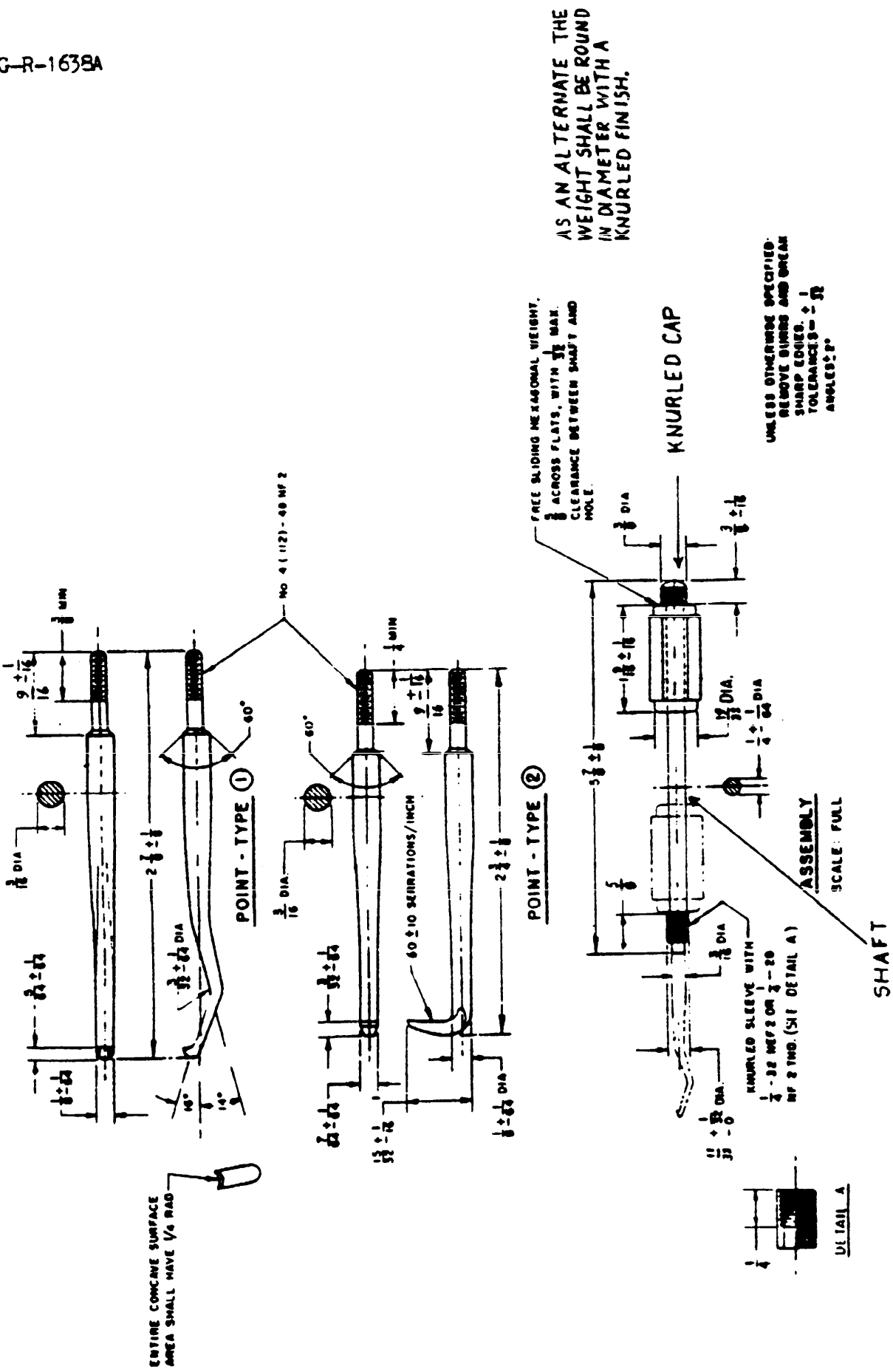


Figure 1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:54Z
Check the source to verify that this is the current version before use.

3.6 Finish. The points shall have a smooth satin or smooth bright finish, except for the sharpened working end which may have a fine honed finish. The remainder of the crown remover shall have a smooth satin or smooth bright finish, except for the knurled portion.

3.7 Corrosion-resistance. The shaft and points shall meet the copper sulfate test requirements of GG-I-526, and the entire assembled crown remover shall meet the boil test requirements of GG-I-526.

3.8 Identification marking. Crown remover shall be permanently and legibly marked in accordance with GG-I-526. Contra-angle points (Type 1) shall be marked with the number "1" and straight angle points (Type 2) shall be marked with the number "2". Each crown remover and each point shall be permanently and legibly marked with the name or registered trademark of the manufacturer (and contractor, if other than manufacturer). Instruments of foreign manufacture shall be permanently and legibly marked with the name of the country of origin, together with the name or registered trademark of the contractor and manufacturer.

3.9 Workmanship. The crown remover shall be free from defects which may detract from its appearance or impair its serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, individual test results, and specification requirements.

4.2 Quality conformance inspection. The examination and tests required to assure conformance of these items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table I. Unit of product for examination purposes shall be one crown remover.

Table I. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for inspection purposes shall be one crown remover.

Table II. Tests

Component	Characteristic	Requirements	Test procedure	Inspection level
Points	Hardness	3.3	GG-I-526	S-2
	Copper Sulfate	3.2.1	GG-I-526	S-2
Entire assembled instrument	Boil test	3.2.1 and 3.2.2	GG-I-526	S-2
	Impact test	3.5	4.4.2	S-2

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the crown remover shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The crown remover shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any crown remover in the sample containing one or more defects shall be rejected, and if the number of defective crown removers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The crown remover shall be examined for defects including, but not limited to, those listed in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

Table III. Classification of Defects

Category	Defects
Major A:	
101	Style and design not as specified.
102	Shaft working surface not free from burrs.
103	Shaft not free from cracks.
104	Points not free from burrs or cracks.
105	Type 1 point (inside face not sharpened).
106	Type 2 points not serrated.
107	Construction not as specified.
Major B:	
151	Dimensions of operative ends of points not as shown on applicable figure.
Minor A:	
201	Identification marking not as specifed.
202	Finish of cap, sleeve, shaft, and weight not as specifed.
203	Finish not free of coarse grind marks.
204	Crown remover not free from dirt, polishing compound, or foreign matter.
205	Packaging and packing not as specified.
206	Marking on packaging and pack not as specified.
Minor B:	
251	Dimensions (other than 151) not as specified.

4.4 Test procedures.

4.4.1 Hardness. The hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E92 (Vickers).

4.4.2 Impact test. The number 2 (straight angle) point shall be securely threaded in the socket of the crown remover. The crown remover shall be securely fixed at a point around the threaded knurled sleeve, or above, in such a manner that the fixation will not interfere with the sliding weight as it moves vertically upward on the shaft, from cap to threaded knurled sleeve. A 20 ± 1 pound weight shall be secured to the sliding weight so as not to interfere with the free movement of the sliding weight. The sliding weight, with its added weight, shall be pushed up the shaft until the sliding weight makes contact with the bottom of the threaded knurled sleeve, and then it shall be released. This impact test shall be performed five times, and then the cap shall be examined for evidence of movement or loosening of cap.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Crown removers shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Crown removers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Intimate wrap. Each crown remover shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.1.5 Unit package. Each protected crown remover shall be packaged in accordance with MIL-P-116, method IC-2. As an alternate, each crown remover shall be packaged in a rigid plastic tube of appropriate size having a friction or plug type closure. The tube shall contain suitable cushioning material to immobilize and protect the item from damage.

5.1.1.6 Intermediate package. Six unit packages shall be packed in a box of appropriate size conforming to PPP-B-566; PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for crown removers shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall be waterproofed as specified in PPP-B-636. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification.

5.2.2 Level B. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for crown removers shall conform to MIL-STD-794 requirements for this level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unititized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Crown removers packed as specified in 5.2.1 and 5.2.2., shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Crown removers packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C.

5.3.1.1 Unit and Intermediate packages. Each unit and each intermediate package shall be marked as specified in MIL-STD-129. In addition, marking shall include the name of the contractor.

5.3.1.2 Exterior container and unitized loads. Exterior containers and unitized loads shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2. Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-551-6150	Remover, Dental Crown, Simplex

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

6.3 Supersession data. This specification includes the requirements of Interim Federal Specification GG-R-001638 (DSA-DM) dated April 19, 1970.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DoD-MB

Agent:
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:
VA-OSS

PROJECT NO. 6520-1825

48146

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Defense Personnel Support Center
Directorate of Medical Materiel
DPSC-ATT
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:54Z
Check the source to verify that this is the current version before use.

(particularly Stock No. and Item Identification)